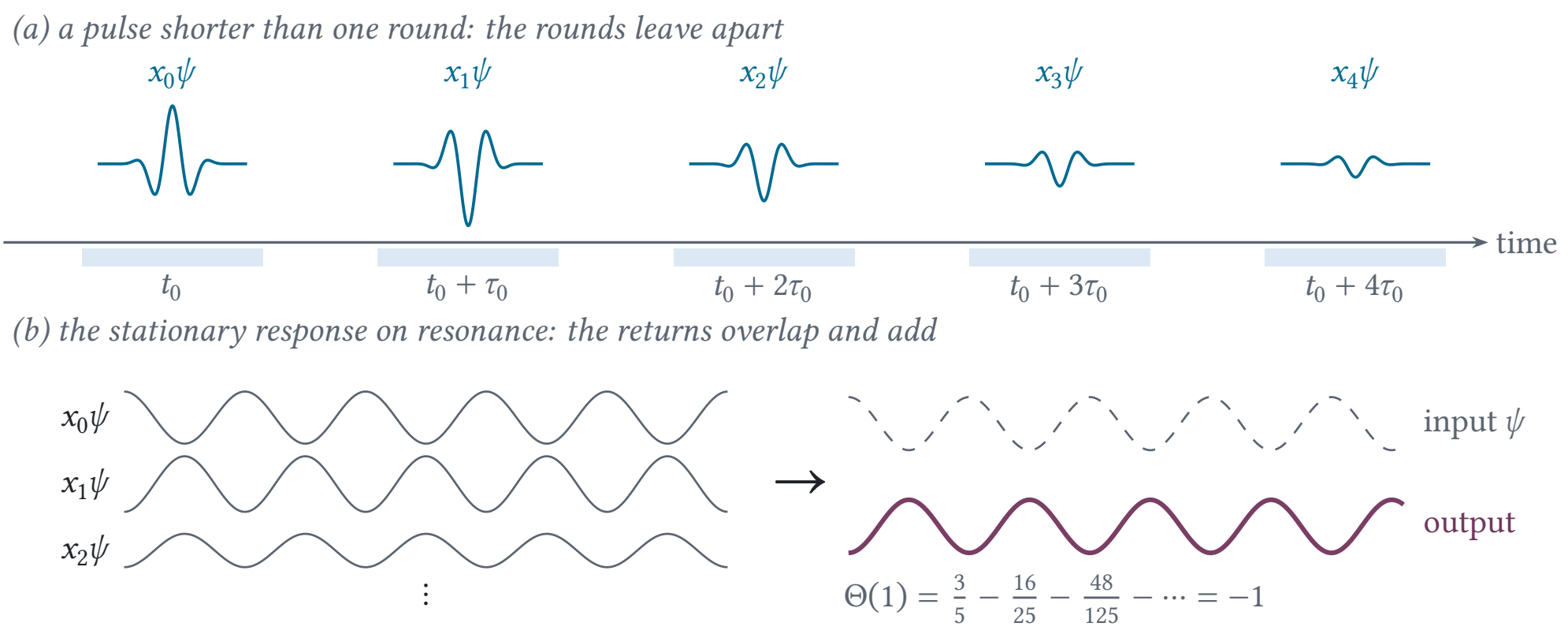
(a) a pulse shorter than one round: the rounds leave apart
$x_0\psi$
$x_1\psi$
$x_2\psi$
$x_3\psi$
$x_4\psi$
time
$t_0$
$t_0 + \tau_0$
$t_0 + 2\tau_0$
$t_0 + 3\tau_0$
$t_0 + 4\tau_0$
(b) the stationary response on resonance: the returns overlap and add
$x_0\psi$
$x_1\psi$
$x_2\psi$
input $\psi$
output
$\Theta(1) = \frac{3}{5} - \frac{16}{25} - \frac{48}{125} - \cdots = -1$